Databases and its Applications

Advanced NoSQL and Emerging Database trends

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Computer Applications

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Aggregation Operators: *sort*, limit, Expressions

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Aggregation Operators in MongoDB

- Aggregation operators are used inside pipeline stages to **process, sort, and transform data**.
- In this topic, we focus on:

    `$sort`

    `$limit`

    Aggregation expressions

Operators in Aggregation Pipeline

- Operators are used **inside stages** such as:
  - `$sort` stage
  - `$limit` stage
  - `$project` and `$group`
- They act on documents passed from previous stages.

$sort Operator

- **$sort** is used to arrange documents in ascending or descending order.
- Sorting is based on one or more fields.
- Order values:

  $1 \rightarrow$ Ascending

  $-1 \rightarrow$ Descending

$sort Example

- Task: Sort students by CGPA in ascending order.

```
db.students.aggregate([ {$sort:{ cgpa:1 }}])
```

- Lowest CGPA appears first.

$sort Example: Descending

- Task: Display top-performing students first.

```
db.students.aggregate([ {$sort:{ cgpa:-1 }}])
```

- Highest CGPA appears first.

Sorting by Multiple Fields

- Sorting can be done using multiple fields.
- Task: Sort by department first, then by CGPA.

```
db.students.aggregate([ {$sort:{ department:1,
cgpa:-1 }}])
```

$limit Operator

- **$limit** restricts the number of documents returned.
- It is commonly used with `$sort`.
- Useful for:

    top-N queries

    pagination

$limit Example

- Task: Display only first 3 students.

```
db.students.aggregate([ {$limit:3} ])
```

- Returns only 3 documents.

$sort + $limit

- Task: Find top 3 students based on CGPA.
  `db.students.aggregate([ {$sort:{ cgpa:-1 }}, {$limit:3} ])`
- Common exam-oriented query.

Aggregation Expressions

- Aggregation expressions perform **calculations** on document fields.
- Used inside:

  `$project`

  `$group`

- Expressions begin with $.

Expression: $sum

- Task: Count number of students per department.

```
db.students.aggregate([ {$group:{
_id:"$department", total:{$sum:1} }}])
```

- `$sum` adds values or counts documents.

Expression: $avg

- Task: Calculate average CGPA per department.

```
db.students.aggregate([ {$group:{
_id:"$department", avgCGPA:{$avg:"$cgpa"} }}])
```

- Used for performance analysis.

Expressions in $project

- Expressions can compute new fields.
- Task: Display CGPA percentage.

```
db.students.aggregate([ {$project:{ name:1,
percentage:{$multiply:["$cgpa",10]} }}])
```

Order of Stages

- Aggregation stages execute sequentially.
- Example order:

```
$match
$sort
$limit
$project
```

- Incorrect order may produce wrong results.

Thank You

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